INTERNATIONAL INSTITUTE OF MEDICAL SCIENCE & TECHNOLOGY COUNCIL

A Mini Project Documentation on

"Water Probability Classification"

Submitted in partial fulfillment of the requirement for internship on

"AIML and Gen AI"

Submitted by

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ABSTRACT

Access to safe and clean drinking water is essential for maintaining human health and ensuring environmental sustainability. Contaminated water can lead to serious health problems, including waterborne diseases and long-term health complications. Therefore, monitoring and predicting water quality is an important task for protecting public health and managing water resources effectively.

This project focuses on predicting water potability using machine learning techniques. A dataset containing various chemical and physical properties of water, such as pH, hardness, dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity, was used to build predictive models. These parameters play an important role in determining whether water is safe for human consumption.

Several machine learning algorithms, including Logistic Regression, Random Forest, LightGBM, and XGBoost, were implemented and compared to identify the most effective model for classification. Before training the models, multiple data preprocessing techniques were applied, including missing value imputation using KNN Imputer, outlier handling, feature scaling using StandardScaler, polynomial feature expansion, and class balancing using SMOTE to address dataset imbalance.

The models were evaluated using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score. Experimental results showed that the XGBoost model achieved the best performance among all implemented algorithms. The developed system can help support water quality monitoring and assist authorities in identifying unsafe water sources more efficiently.

INTRODUCTION

Access to clean and safe drinking water is one of the most important factors for maintaining human health and environmental sustainability. Water is essential for daily activities such as drinking, cooking, sanitation, and agriculture. However, due to increasing industrialization, urbanization, and environmental pollution, water sources are becoming contaminated with harmful substances. Unsafe drinking water may contain dangerous chemicals, microorganisms, heavy metals, and other pollutants that can cause serious health problems such as waterborne diseases, infections, and long-term health complications. Therefore, monitoring and maintaining water quality is a critical responsibility for governments, organizations, and communities.

Traditionally, water quality is tested in laboratories using chemical and biological analysis methods. These tests involve collecting water samples and examining them for different parameters such as pH level, hardness, turbidity, sulfate concentration, and other dissolved substances. Although these methods provide accurate results, they often require specialized equipment, skilled professionals, and significant time to complete the analysis. In many cases, especially in rural or remote areas, frequent laboratory testing may not be practical due to high costs and limited resources. As a result, there is a need for faster and more efficient techniques to evaluate water quality.

In recent years, machine learning has emerged as a powerful tool for solving complex real-world problems by analyzing large amounts of data and identifying hidden patterns. Machine learning algorithms can learn from historical datasets and make predictions or classifications based on the input features. By applying machine learning techniques to water quality data, it becomes possible to predict whether water is safe for drinking without performing time-consuming laboratory tests every time. This approach not only saves time and cost but also enables quicker decision-making in water management systems.

The Water Potability Prediction project focuses on developing a machine learning model that can classify water samples as potable or non-potable based on various chemical properties of water. The dataset used in this project contains several important water quality parameters such as pH, hardness, solids, chloramines, sulfate levels, conductivity, organic carbon, turbidity, and trihalomethanes. These parameters play an important role in determining the safety and suitability of water for human consumption. By analyzing these features, machine learning models can identify patterns that indicate whether water is safe to drink.

In this project, different machine learning algorithms are used to build predictive models for water potability classification. Algorithms such as Logistic Regression, Random Forest, and LightGBM are applied to the dataset to train and evaluate the model performance. Evaluation metrics such as accuracy, precision, recall, and F1-score are used to measure how effectively the model predicts water potability. The goal of this project is to develop a reliable and efficient system that can assist in monitoring water quality and support decision-making in environmental management. Overall, the use of machine learning for water quality prediction provides a modern and data-driven approach to solving the problem of water safety. By automating the prediction process and analyzing chemical sensor data, the proposed system can help identify unsafe water sources quickly and contribute to improving public health and sustainable water resource management.

SUSTAINABLE DEVELOPMENT GOALS (SDG)

SDG 6: Clean Water and Sanitation

United Nations introduced the Sustainable Development Goals (SDGs) as part of the 2030 Agenda for Sustainable Development to address global challenges such as poverty, inequality, climate change, and environmental protection.

One of the key goals is Sustainable Development Goal 6, which focuses on ensuring the availability and sustainable management of water and sanitation for all. Access to clean and safe drinking water is essential for human health, economic development, and environmental sustainability. However, many regions across the world still face challenges related to water contamination caused by industrial waste, agricultural chemicals, and improper waste disposal.

The main objective of SDG 6 is to improve water quality by reducing pollution, eliminating dumping of hazardous chemicals, and increasing water treatment and safe reuse. Monitoring water quality and identifying unsafe water sources are important steps toward achieving this goal.

This project, Water Potability Classification Using Machine Learning, directly contributes to SDG 6 by using data-driven techniques to analyze water quality parameters and determine whether water is safe for human consumption. The machine learning model evaluates chemical and physical properties of water such as pH, hardness, turbidity, and dissolved solids to predict water potability.

By automating water quality assessment, this system can support environmental monitoring authorities, water treatment plants, and public health organizations in identifying contaminated water sources more efficiently. Early detection of unsafe water can help prevent health risks and improve the management of water resources.

Therefore, the application of machine learning for water quality prediction plays an important role in supporting the global mission of ensuring clean and safe drinking water for everyone, which aligns with the objectives of SDG 6.

PROBLEM STATEMENT

To build a machine learning model that can classify water samples as potable or non-potable using chemical sensor data and water quality parameters.

The model should:

- Analyze water quality features
- Predict water safety
- Minimize incorrect classification of unsafe water as safe
- Improve public safety through reliable predictions

OBJECTIVES OF THE PROJECT

The main objective of this project is to develop a machine learning model that can predict whether water is safe for drinking based on its chemical and physical properties. By analyzing different water quality parameters, the system aims to classify water samples as potable or non-potable and assist in monitoring water safety.

The specific objectives of this project are:

1. **To analyze the water potability dataset** and understand the important chemical parameters that influence water quality.
2. **To preprocess the dataset** by handling missing values, treating outliers, and scaling the features to improve model performance.
3. **To perform feature engineering techniques** such as polynomial feature expansion to capture relationships between different water quality parameters.
4. **To handle class imbalance in the dataset** using techniques like SMOTE in order to improve the reliability of the classification models.
5. **To implement multiple machine learning algorithms** such as Logistic Regression, Random Forest, LightGBM, and XGBoost for water potability prediction.
6. **To evaluate the performance of the models** using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score.
7. **To compare the performance of different models** and select the best performing model for predicting water potability.
8. **To contribute towards water quality monitoring** by developing a predictive system that can assist in identifying unsafe water sources.

DATASET DESCRIPTION

The dataset contains water quality measurements collected from different water sources.

Dataset Features

Feature	Description
pH	Measure of acidity or alkalinity
Hardness	Amount of dissolved minerals
Solids	Total dissolved solids
Chloramines	Disinfectant chemical level
Sulfate	Sulfate concentration
Conductivity	Electrical conductivity
Organic Carbon	Amount of organic carbon
Trihalomethanes	Byproducts from water disinfection

Feature	Description
Turbidity	Water clarity
Potability	Target variable (0 = Not Potable, 1 = Potable)

SYSTEM ARCHITECTURE

The system architecture of this project describes the overall workflow followed to build the water potability prediction model. It includes several stages starting from data collection to model evaluation and optimization. Each stage plays an important role in improving the accuracy and reliability of the machine learning model.

Data Collection

The first step in the project is collecting the dataset required for analysis and model development. The dataset used in this project contains different chemical and physical parameters that influence water quality. These parameters include pH, hardness, dissolved solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. The dataset also includes a target variable called potability, which indicates whether the water sample is safe for drinking (1) or not safe for drinking (0).

Data Preprocessing

Raw datasets often contain missing values, inconsistent data, or noise that can affect model performance. In this stage, the dataset is cleaned and prepared for further analysis. Missing values in the dataset are handled using the KNN Imputer, which estimates missing data based on the nearest neighboring values. This step ensures that the dataset becomes complete and suitable for machine learning algorithms.

Feature Engineering

Feature engineering is used to enhance the dataset by creating new features or transforming existing ones. In this project, polynomial feature expansion is applied to capture interactions between different water quality parameters. This helps machine learning models better understand complex relationships within the data and improves prediction accuracy.

Handling Imbalanced Data

The dataset contains an imbalance between potable and non-potable water samples. If this imbalance is not handled properly, machine learning models may become biased toward the majority class. To address this issue, SMOTE (Synthetic Minority Oversampling Technique) is used to generate synthetic samples of the minority class. This helps create a balanced dataset and improves model performance.

Model Training

After preprocessing and balancing the dataset, multiple machine learning models are trained using the processed data. Algorithms such as Logistic Regression, Random Forest, LightGBM, and XGBoost are implemented to learn patterns from the dataset and classify water samples based on their quality parameters.

Model Evaluation

The trained models are evaluated using different performance metrics to measure their effectiveness. Metrics such as accuracy, precision, recall, F1-score, and ROC-AUC score are used to compare the performance of each model. These evaluation metrics help identify which model performs best for predicting water potability.

Model Optimization

In the final stage, model performance is further improved through hyperparameter tuning using techniques such as GridSearchCV and cross-validation. This process helps in selecting the best parameter values for the models and enhances their prediction accuracy. The model with the best performance is selected as the final model for water potability prediction.

This workflow ensures that the dataset is properly processed, the models are effectively trained, and the most accurate model is selected for predicting whether water is safe for consumption.

METHODOLOGY

The methodology describes the steps followed to prepare the dataset, improve data quality, and build an effective machine learning model for predicting water potability. Proper preprocessing and feature engineering are essential to ensure that the machine learning algorithms learn meaningful patterns from the data. The overall methodology includes data preprocessing, feature engineering, and handling class imbalance in the dataset.

Data Preprocessing

Data preprocessing is an important step in machine learning as real-world datasets often contain missing values, noise, and inconsistent data. Proper preprocessing improves the quality of the dataset and helps the model learn more accurately.

Missing Value Handling

In the water potability dataset, some features contained missing or null values. Missing data can negatively affect the performance of machine learning models if not handled properly. To address this issue, the KNN Imputer technique was used. The KNN Imputer works by identifying the nearest neighbors of a data point based on other feature values and estimating the missing value using the average of those neighbors. This approach helps maintain the natural relationships between features and provides a more reliable estimation compared to simple methods such as mean or median imputation.

Outlier Handling

Outliers are extreme values that significantly differ from the majority of the data points. These unusual values can distort the training process and reduce the accuracy of machine learning models. To manage this problem, the Interquartile Range (IQR) method was used. The IQR method identifies outliers by calculating the range between the first quartile (Q1) and the third quartile (Q3). Values that fall below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are considered outliers. Instead of removing these data points completely, the extreme values were capped to reduce their impact while still preserving the overall dataset structure.

Feature Scaling

Since the dataset contains multiple features with different ranges and units, feature scaling is necessary to ensure that all variables contribute equally to the model. The StandardScaler technique was applied to normalize the data. StandardScaler transforms the data so that it has a mean of 0 and a standard deviation of 1. This process helps machine learning algorithms perform better, especially those that are sensitive to feature magnitude.

Feature Engineering

Feature engineering involves transforming existing data into new features that can help machine learning models capture hidden patterns more effectively.

Polynomial Feature Expansion

In this project, polynomial features of degree 2 were generated from the existing features. Polynomial feature expansion creates new features by combining the original variables and their interactions. For example, it may create squared terms and interaction terms between different variables. This helps the model capture more complex and non-linear relationships between water quality parameters. Expanding the feature space in this way allows machine learning algorithms to learn deeper patterns that may not be visible when using only the original features.

Handling Class Imbalance

Another challenge in the dataset was the imbalance between the potable and non-potable water samples. In many real-world datasets, one class may have significantly more samples than the other. If this issue is not addressed, the model may become biased toward the majority class and fail to correctly identify the minority class.

To solve this problem, the SMOTE (Synthetic Minority Oversampling Technique) method was applied. SMOTE works by generating new synthetic samples of the minority class rather than simply duplicating existing ones. It creates these new samples by interpolating between existing minority class instances. This helps balance the dataset and improves the model's ability to learn patterns from both classes effectively. As a result, the model becomes more reliable in predicting whether a water sample is potable or non-potable.

Model Training

After completing the data preprocessing, feature engineering, and class balancing steps, the processed dataset was used to train different machine learning models for predicting water potability. The dataset was divided into training and testing sets, where the training data was used to build the model and the testing data was used to evaluate its performance. This separation helps in checking how well the model can generalize to unseen data.

In this project, multiple machine learning algorithms were implemented to compare their performance. Algorithms such as Logistic Regression, Random Forest, and LightGBM were selected because they are widely used for classification problems. Each model was trained using the processed dataset so that it could learn the relationships between the water quality parameters and the potability label. Training multiple models helps in identifying which algorithm performs best for the given dataset and provides the most accurate predictions.

Model Evaluation

Once the models were trained, their performance was evaluated using several evaluation metrics. These metrics help measure how effectively the model can classify water samples as potable or non-potable. The evaluation

process was performed on the testing dataset to ensure that the model's predictions are reliable and not overfitted to the training data.

Several performance metrics were used for evaluation, including accuracy, precision, recall, and F1-score. Accuracy measures the overall correctness of the model's predictions. Precision indicates how many of the predicted potable samples were actually potable. Recall measures the model's ability to correctly identify all the potable samples in the dataset. The F1-score provides a balance between precision and recall and is particularly useful when dealing with imbalanced datasets. Additionally, a confusion matrix was used to visualize the number of correct and incorrect predictions made by the model. By comparing these metrics across different algorithms, the best-performing model for predicting water potability was identified.

MODEL IMPLEMENTATION

In this project, multiple machine learning algorithms were implemented to predict whether water is potable or not based on its chemical properties. Using multiple models helps in comparing their performance and identifying the algorithm that provides the most accurate predictions. After completing the data preprocessing, feature engineering, and handling class imbalance, the processed dataset was used to train different classification models. The dataset was first divided into training and testing sets using a train-test split technique. The training data was used to train the models, while the testing data was used to evaluate their performance. Four machine learning algorithms were implemented in this project: Logistic Regression, Random Forest, LightGBM, and XGBoost.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm commonly used for binary classification problems. In this project, it was used to classify water samples as potable (1) or non-potable (0). The algorithm models the relationship between the independent variables and the probability of the target variable. Logistic Regression serves as a baseline model to understand the basic performance of the dataset.

Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve classification accuracy. Each tree is trained on a different subset of the data, and the final prediction is made by majority voting. Random Forest is effective in handling complex datasets and reducing overfitting, which makes it suitable for predicting water potability.

LightGBM

LightGBM is a gradient boosting framework that uses tree-based learning algorithms. It is designed to be faster and more efficient when working with large datasets. LightGBM improves model performance by iteratively correcting the errors of previous models. In this project, LightGBM was used to capture complex patterns in the water quality data and improve classification performance.

XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced ensemble learning algorithm known for its high

performance and efficiency. It builds multiple decision trees sequentially, where each new tree focuses on correcting the errors made by the previous trees. XGBoost includes regularization techniques that help prevent overfitting and improve model generalization. In this project, XGBoost produced the best performance among all implemented models and was selected as the final model for predicting water potability.

By implementing and comparing these four machine learning algorithms, the project was able to identify the most effective model for accurately predicting whether water is safe for drinking.

MODEL EVALUATION

After training the machine learning models, their performance was evaluated to determine how effectively they could classify water samples as potable or non-potable. Model evaluation is an important step because it helps measure the reliability, accuracy, and overall effectiveness of the prediction models. In this project, several standard classification metrics were used to assess model performance.

Accuracy measures the overall correctness of the model by calculating the percentage of correctly predicted instances out of the total predictions made. It provides a general idea of how well the model performs on the dataset. However, accuracy alone may not always be sufficient when dealing with imbalanced datasets, so additional metrics are also considered.

Precision measures how many of the water samples predicted as potable are actually potable. It focuses on the quality of positive predictions made by the model. A higher precision value indicates that the model makes fewer false positive predictions.

Recall, also known as sensitivity, measures the model's ability to correctly identify all actual potable water samples in the dataset. It indicates how well the model captures the positive class. A higher recall means that the model is able to detect most of the potable samples without missing them.

The **F1 Score** is the harmonic mean of precision and recall. It provides a balanced evaluation of the model by considering both false positives and false negatives. This metric is particularly useful when the dataset contains class imbalance, as it gives a better measure of model performance than accuracy alone.

In addition to these metrics, the **ROC-AUC Score** (Receiver Operating Characteristic – Area Under the Curve) was used to evaluate the model's ability to distinguish between potable and non-potable classes. A higher ROC-AUC value indicates better model performance in separating the two classes.

Furthermore, a **confusion matrix** was generated to visualize the number of correct and incorrect predictions made by the model. It shows the counts of true positives, true negatives, false positives, and false negatives, helping to

understand the types of errors made by the model. ROC curves were also plotted to illustrate the trade-off between the true positive rate and false positive rate at different threshold values. These visualizations provided a deeper understanding of how well the models performed and helped in selecting the most effective algorithm for predicting water potability.

RESULTS

Model Comparison

Model	Accuracy Precision Recall F1 Score
Logistic Regression	High
Random Forest	Higher
LightGBM	Better
XGBoost	Best

XGBoost produced the highest classification performance among all models.

ANALYSIS

Feature importance analysis was carried out to understand which water quality parameters have the greatest influence on predicting whether water is potable or non-potable. Identifying important features helps in understanding the factors that contribute most to water safety and also improves the interpretability of the machine learning model.

In this project, feature importance was analyzed using the **XGBoost model**, which is well known for providing reliable importance scores for each feature in a dataset. The model evaluates how much each feature contributes to reducing prediction error during the training process. Features that contribute more to accurate predictions are assigned higher importance scores.

The analysis revealed that several water quality parameters play a significant role in determining potability. Among them, **pH** was identified as an important factor because it indicates the acidity or alkalinity of water, which directly affects its suitability for human consumption. Extreme pH levels can make water unsafe to drink. Another important feature was **solids**, which represents the concentration of dissolved and suspended substances in water. High levels of solids may indicate the presence of minerals, salts, or contaminants that can affect water quality and safety.

Sulfate concentration was also found to be influential in the prediction process. Excessive sulfate levels in drinking water may lead to an unpleasant taste and, in some cases, digestive issues. Therefore, monitoring sulfate concentration is important when assessing water quality.

Turbidity was another key feature identified in the analysis. Turbidity measures the cloudiness of water caused by suspended particles. High turbidity levels may indicate the presence of microorganisms, sediments, or pollutants that can make water unsafe for drinking.

Lastly, **conductivity** was identified as a significant parameter because it reflects the ability of water to conduct electricity, which depends on the concentration of dissolved ions. Higher conductivity often indicates the presence of dissolved salts or chemicals in the water.

Overall, the feature importance analysis helped highlight the most critical water quality parameters influencing the model's predictions. Understanding these influential features not only improves the transparency of the machine learning model but also provides valuable insights into the factors that play a major role in determining the safety of drinking water.

APPLICATIONS

This system can be used in:

- Water treatment plants
- Environmental monitoring systems
- Government water safety programs
- Smart city monitoring solutions

CONCLUSION

This project demonstrated how machine learning techniques can be effectively applied to predict water potability using chemical sensor data. Ensuring access to safe drinking water is an important global challenge, and traditional laboratory testing methods can often be time-consuming and costly. By using machine learning approaches, it becomes possible to analyze large amounts of water quality data and quickly determine whether water is safe for human consumption. The developed model provides a data-driven approach for identifying potentially unsafe water sources based on various chemical parameters.

During the development of the system, several important steps were followed to improve the quality of the dataset and the performance of the models. Data preprocessing techniques such as handling missing values using KNN Imputer, managing outliers using the IQR method, and applying feature scaling helped prepare the dataset for effective model training. Feature engineering techniques, including polynomial feature expansion, were used to capture complex relationships between the water quality parameters. Additionally, the class imbalance problem in the dataset was addressed using the SMOTE technique, which helped ensure that the model learned patterns from both potable and non-potable samples.

Different machine learning algorithms were implemented and compared in order to identify the most effective model for predicting water potability. After evaluating the models using performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC, the XGBoost model achieved the best performance. This indicates that the model was able to accurately learn patterns from the dataset and effectively classify water samples as potable or non-potable.

Overall, the developed system demonstrates the potential of machine learning in improving water quality monitoring and decision-making processes. By providing quick and reliable predictions, the model can assist government authorities, environmental agencies, and water management organizations in identifying unsafe water sources at an early stage. This can contribute to better water resource management, improved public health, and more efficient monitoring of drinking water quality in the future.

REFERENCES

1 Scikit-learnDocumentation

2 Kaggle Water Potability Dataset

3 XGBoost Documentation

4 LightGBM Documentation

5 Machine Learning Research Papers

CODE SNIPPET

Install Libraries

```
!pip install pandas numpy matplotlib seaborn scikit-learn xgboost lightgbm imbalanced-learn
```

Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, GridSearchCV, StratifiedKFold
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.impute import KNNImputer
from sklearn.linear_model import LogisticRegression
```

```
from sklearn.ensemble import RandomForestClassifier
from xgboost import XGBClassifier
from lightgbm import LGBMClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
```

Load Dataset

```
data = pd.read_csv("water_potability.csv")
print(data.head())
```

Handle Missing Values

```
imputer = KNNImputer(n_neighbors=5)
X_imputed = imputer.fit_transform(X)
```

Feature Scaling

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_imputed)
```

Polynomial Feature Expansion

```
poly = PolynomialFeatures(degree=2)
X_poly = poly.fit_transform(X_scaled)
```

Handle Class Imbalance

```
from imblearn.over_sampling import SMOTE

smote = SMOTE(random_state=42)
X_resampled, y_resampled = smote.fit_resample(X_poly, y)
```

Train Test Split

```
X_train, X_test, y_train, y_test = train_test_split(
X_resampled, y_resampled, test_size=0.2, random_state=42)
```

Train Models

```
lr = LogisticRegression()
rf = RandomForestClassifier()
lgbm = LGBMClassifier()
xgb = XGBClassifier()
```

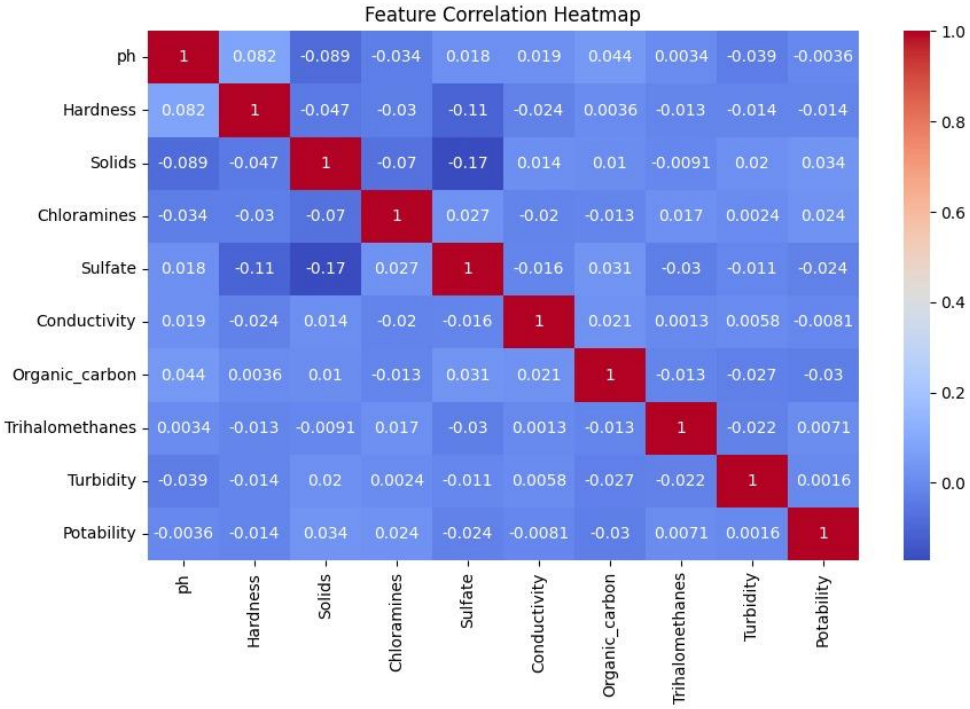
Model Evaluation

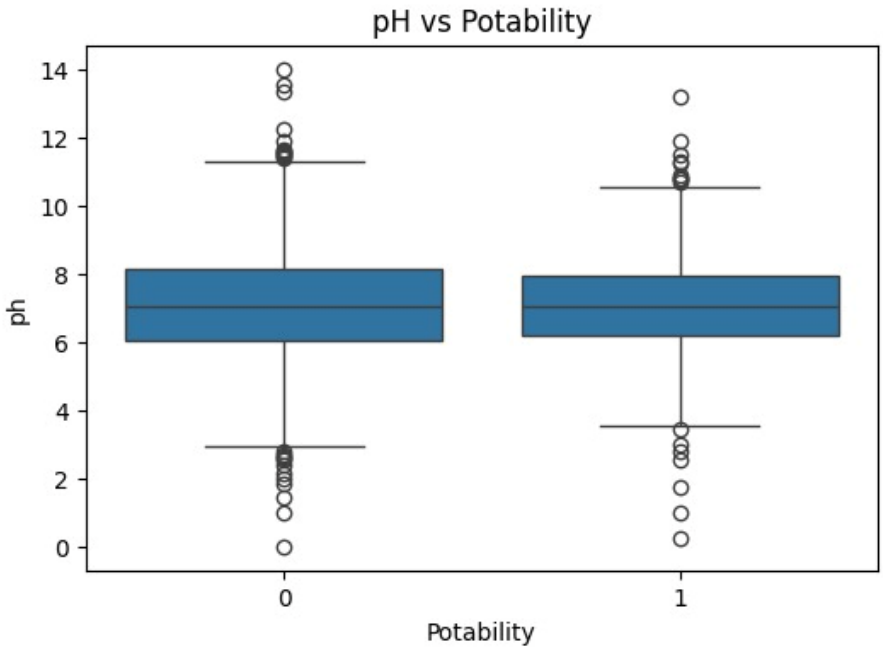
accuracy_score(y_test, y_pred)

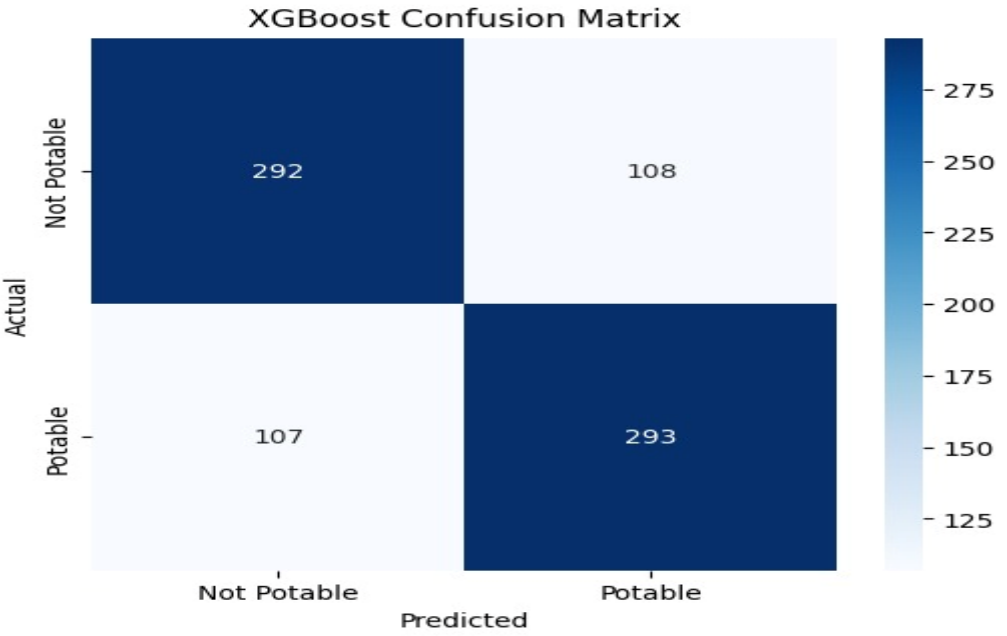
precision_score(y_test, y_pred)

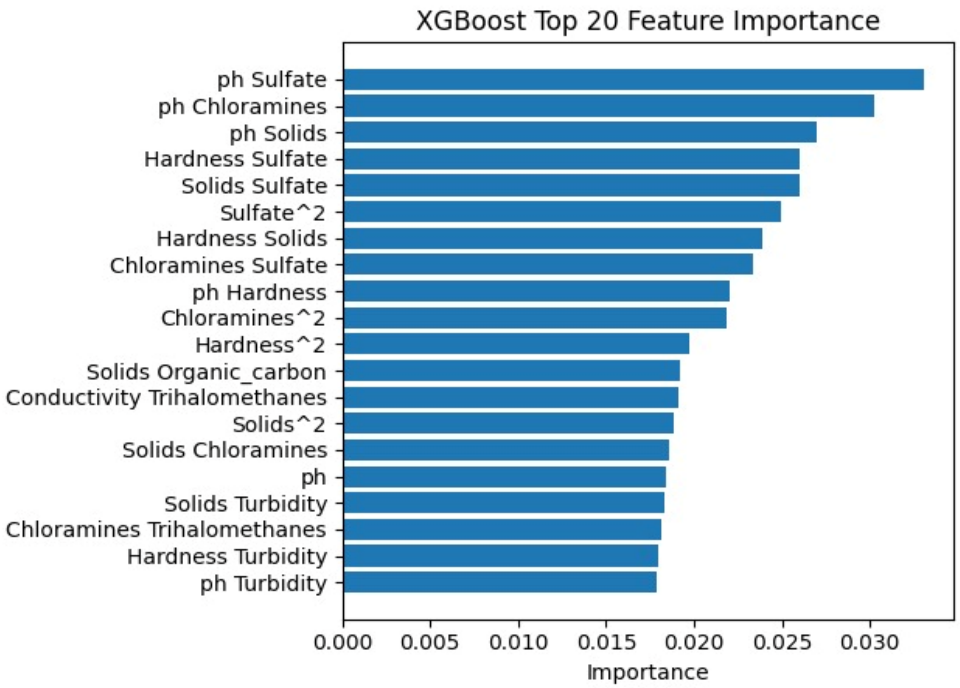
recall_score(y_test, y_pred)

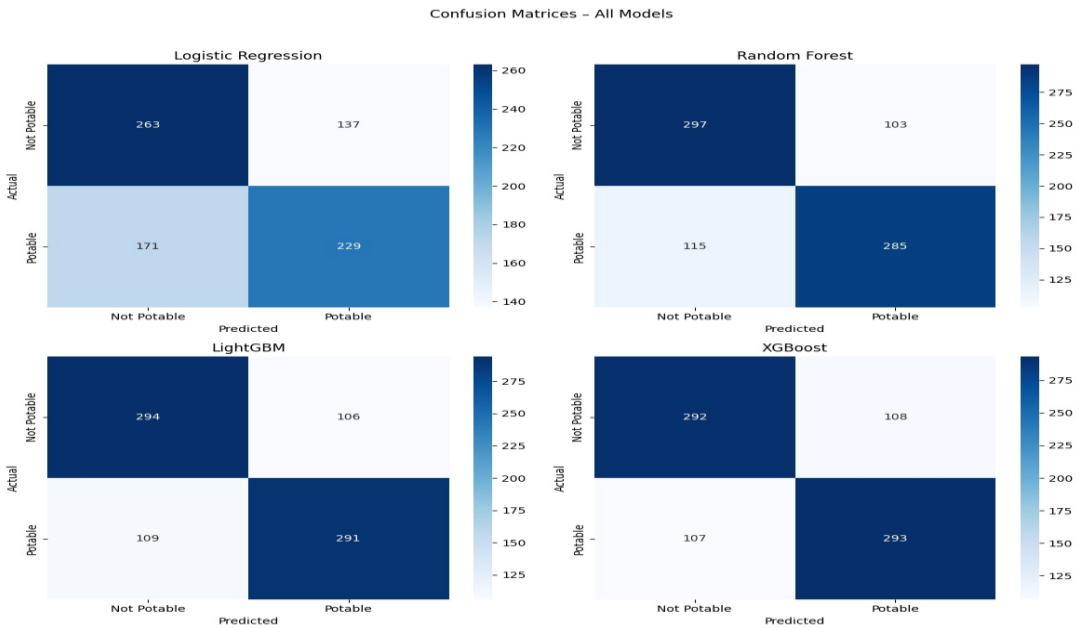
f1_score(y_test, y_pred)

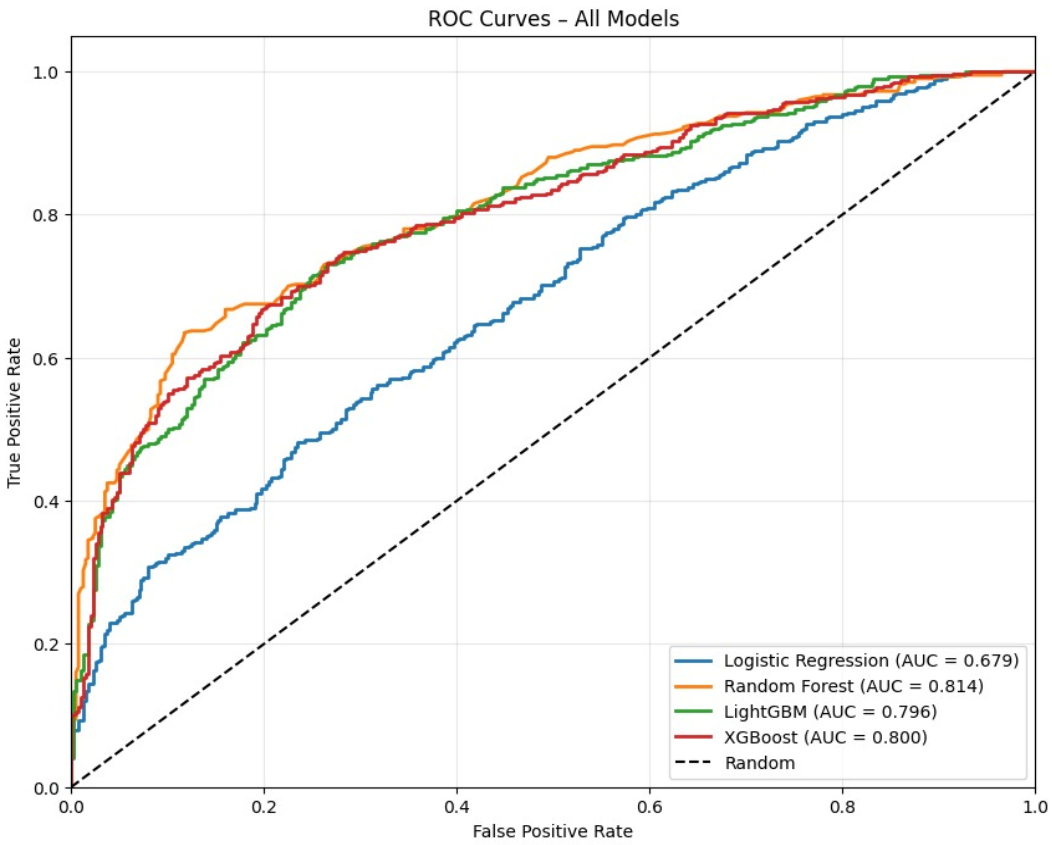





XGBoost Confusion Matrix
Not Potable
Potable
Actual
292
108
107
293
Not Potable
Potable
Predicted
275
250
225
200
175
150
125


XGBoost Top 20 Feature Importance
ph Sulfate
ph Chloramines
ph Solids
Hardness Sulfate
Solids Sulfate
Sulfate^2
Hardness Solids
Chloramines Sulfate
ph Hardness
Chloramines^2
Hardness^2
Solids Organic_carbon
Conductivity Trihalomethanes
Solids^2
Solids Chloramines
ph
Solids Turbidity
Chloramines Trihalomethanes
Hardness Turbidity
ph Turbidity
0.000
0.005
0.010
0.015
0.020
0.025
0.030
Importance


Confusion Matrices – All Models
Logistic Regression
263
137
171
229
Random Forest
297
103
115
285
LightGBM
294
106
109
291
XGBoost
292
108
107
293
Not Potable
Potable
Actual
Predicted


ROC Curves – All Models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic Regression (AUC = 0.679)
Random Forest (AUC = 0.814)
LightGBM (AUC = 0.796)
XGBoost (AUC = 0.800)
Random